

Unbounded Shell-Multiplicity Pathology

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Abstract

We relax the RH-248 single-use shell zonotope to the full nonnegative cone. Six of 32 frozen endpoints remain outside the anchored tolerance, with primal–dual certificates. The other 26 can be fitted only by shell weights whose minimum necessary maxima range from 40.58 to 5.80×10^{10} . Such arbitrary real multiplicities are not a spectral cloud. The formal success is therefore a moment-reweighting pathology, not coefficient identification.

1 Cone relaxation and dual

Retain the real shell power matrix V , target difference d , and weights $\omega_n = 1/n$ from RH-248 [3]. Remove the upper bounds on shell use and define

$$\delta_+ = \min_{w \geq 0} \sum_{n=2}^{12} \omega_n |d_n - (Vw)_n|. \quad (1)$$

Theorem 1 (nonnegative-cone dual). *The cone distance satisfies*

$$\delta_+ = \max_{\substack{|y_n| \leq \omega_n \\ V^T y \leq 0}} y^T d. \quad (2)$$

Thus any feasible dual vector with value greater than σ excludes all nonnegative shell weights, regardless of their size.

Proof. Dualize the weighted ℓ^1 residual as in RH-248. The infimum of $-y^T Vw$ over $w \geq 0$ is finite exactly when $V^T y \leq 0$, in which case it is zero. Finite-dimensional LP strong duality gives (2). $\square$

For endpoints with $\delta_+ \leq \sigma$, a second LP quantifies the pathology:

$$W_* = \min \{W : 0 \leq w_j \leq W, \sum_n \omega_n |d_n - (Vw)_n| \leq \sigma\}. \quad (3)$$

This is the smallest possible maximum shell weight, not the weight of an arbitrary optimal cone solution.

2 Finite result

The frozen RH-222 windows and RH-243 anchor give 26 cone passes and six failures [1, 2]. The failing endpoints are

$$(0.04, L), (0.02, L), (0.02, R), (0.016, R), (0.00625, L), (0.00625, R).$$

Their smallest distance-minus-tolerance margin is $2.826242139956541 \times 10^{-4}$, while the maximum recorded primal–dual gap is 4.08×10^{-7} . Hence the finite separation is much larger than the LP consistency error.

For the 26 formal passes, solving (3) gives

$$40.58443731031147 \leq W_* \leq 58018432630.629776.$$

Seventeen cone fits can drive the order-12 jet objective to floating zero, yet may still require thousands to billions of repetitions. With the common cap $W = 40$, no endpoint passes; at $W = 41$, exactly one endpoint passes.

quantity	value
unbounded cone passes/failures	26 / 6
minimum required cap among passes	40.5844373
maximum required cap among passes	5.8018433×10^{10}
passes with cap 40	0/32
passes with cap 41	1/32

3 Interpretation and boundary

An algebraic spectral multiset has fixed nonnegative integer multiplicities. The variables in (1) are arbitrary real repetitions of already resolved shells, often enormous. They alter the moment problem without constructing a corresponding noisy operator or invariant root space. The 26 LP passes therefore do not supply the RH-243 coefficient bridge.

The six failures and multiplicity explosion concern only the frozen candidate windows and orders 2–12. They do not exclude expanded windows, deeper roots, signed/complex quotient grouping, or continuum mechanisms. No all-order envelope follows. Gate A remains open and Gates B–E are untouched; no Hilbert–Polya operator, zeta-divisor equality, Riemann-zero identification, or RH implication is claimed.

References

- [1] Bin Wang. Rank-growing conjugate cloud atlas. 2026. RH-222 preprint.
- [2] Bin Wang. Deterministic numerator coefficient anchor dictionary. 2026. RH-243 preprint.
- [3] Bin Wang. Anchored shell-zonotope reachability obstruction. 2026. RH-248 preprint.